

2009 H2 Maths Prelim(Paper 2) Solutions

1 $\frac{dx}{dt} = \frac{a}{x} - bx$ where a and b are positive constants

When $x = 2$, $\frac{dx}{dt} = 0 \Rightarrow \frac{a}{2} = b(2) \Rightarrow a = 4b$

$\therefore \frac{dx}{dt} = \frac{4b}{x} - bx$

$$= -b \left(\frac{x^2 - 4}{x} \right)$$

$$= k \left(\frac{x^2 - 4}{x} \right), \text{ where } k \text{ is a negative constant}$$

$$\int \frac{x}{x^2 - 4} dx = \int k dt$$

$$\frac{1}{2} \ln |x^2 - 4| = kt + C$$

Since $x > 2$, $x^2 - 4 = e^{2C} e^{2kt}$

$$x^2 = Ae^{2kt} + 4 \quad \text{where } A = e^{2C}$$

$$x = \sqrt{(Ae^{2kt} + 4)} \quad (\text{Reject } -\sqrt{(Ae^{2kt} + 4)} \text{ since } x > 0)$$

2(i) By similar triangles, $\frac{x}{3} = \frac{6-h}{6}$

$$x = \frac{1}{2}(6-h)$$

$$V = \frac{1}{3}\pi x^2 h$$

$$V = \frac{1}{3}\pi \left(\frac{1}{2}\right)^2 (6-h)^2 h$$

$$V = \frac{1}{12}\pi (h^3 + 36h - 12h^2) \quad (\text{Shown})$$

2(ii) Let $\frac{dV}{dh} = 0$

$$\Rightarrow \frac{1}{12}\pi (3h^2 - 24h + 36) = 0$$

$$\Rightarrow h^2 - 8h + 12 = 0$$

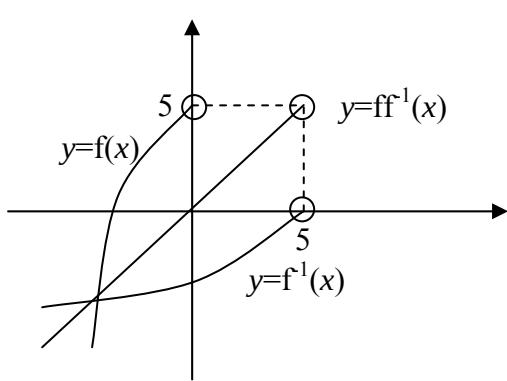
$$\Rightarrow h = 2 \quad \text{or} \quad h = 6 \quad (\text{NA as } V = 0 \text{ when } h = 6)$$

$$\frac{d^2V}{dh^2} = \frac{\pi}{2}(h - 4)$$

$$\left. \frac{d^2V}{dh^2} \right|_{h=2} < 0$$

Therefore the maximum volume of the inverted cone occurs when $h = 2$.

	$\text{Max. } V = \frac{1}{3} \pi \left(\frac{1}{2} \right)^2 (6-h)^2 h = 8.38 \text{ m}^3$
3(i)	$ \begin{aligned} u_{r-1} - u_{r+1} &= \frac{1}{(r-1)^3} - \frac{1}{(r+1)^3} \\ &= \frac{(r+1)^3 - (r-1)^3}{[(r-1)(r+1)]^3} \\ &= \frac{(r^3 + 3r^2 + 3r + 1) - (r^3 - 3r^2 + 3r - 1)}{(r^2 - 1)^3} \\ &= \frac{6r^2 + 2}{(r^2 - 1)^3} \text{ (shown)} \end{aligned} $
3(ii)	$ \begin{aligned} \sum_{r=2}^n \frac{3r^2 + 1}{(r^2 - 1)^3} &= \frac{1}{2} \sum_{r=2}^n \frac{6r^2 + 2}{(r^2 - 1)^3} \\ &= \frac{1}{2} \sum_{r=2}^n [u_{r-1} - u_{r+1}] \\ &= \frac{1}{2} \begin{bmatrix} +u_1 & \cancel{-u_3} \\ +u_2 & \cancel{-u_4} \\ \cancel{+u_3} & \cancel{-u_5} \\ \vdots & \vdots \\ \cancel{+u_{n-3}} & \cancel{-u_{n-1}} \\ \cancel{+u_{n-2}} & -u_n \\ \cancel{+u_{n-1}} & -u_{n+1} \end{bmatrix} \\ &= \frac{1}{2} [u_1 + u_2 - u_n - u_{n+1}] \\ &= \frac{1}{2} \left[1 + \frac{1}{8} - \frac{1}{n^3} - \frac{1}{(n+1)^3} \right] \\ &= \frac{1}{2} \left[\frac{9}{8} - \frac{1}{n^3} - \frac{1}{(n+1)^3} \right] \end{aligned} $

3(iii)	$\sum_{r=3}^n \frac{3(r-1)^2 + 1}{[(r-1)^2 - 1]^3}$ $= \sum_{s=2}^{n-1} \frac{3s^2 + 1}{(s^2 - 1)^3} \quad \text{Subt. } r-1 = s. \text{ When } r=3, s=2. \text{ When } r=n, s=n-1$ $= \frac{1}{2} \left[\frac{9}{8} - \frac{1}{(n-1)^3} - \frac{1}{n^3} \right]$ As $n \rightarrow \infty$, $\frac{1}{n^3} \rightarrow 0$ and $\frac{1}{(n-1)^3} \rightarrow 0$. $\sum_{r=3}^{\infty} \frac{3(r-1)^2 + 1}{[(r-1)^2 - 1]^3} = \frac{1}{2} \left[\frac{9}{8} - 0 - 0 \right] = \frac{9}{16}$
4	(i) Largest $k = 0$ Let $y = 5 - x^2$ $x = -\sqrt{5-y}$ ($x = \sqrt{5-y}$ is rejected as $x < 0$) $f^{-1} : x \mapsto -\sqrt{5-x}$, $x < 5$ (ii)  $f(x) = f^{-1}(x) \Rightarrow f(x) = x$ $5 - x^2 = x$ $x^2 + x - 5 = 0$ $x = \frac{-1 \pm \sqrt{21}}{2}$ (Rej $x = \frac{-1 + \sqrt{21}}{2}$) $\therefore x = \frac{-1 - \sqrt{21}}{2}$ (ans in exact form) (iii) $R_{f^{-1}} = D_f = (-\infty, 0)$ $D_g = (-\infty, \infty)$ Since $R_{f^{-1}} \subseteq D_g$, gf^{-1} exists.

	$R_g = [0, \infty)$ $D_{f^{-1}} = (-\infty, 5)$ Since $R_g \not\subset D_{f^{-1}}$, $f^{-1}g$ does not exist.
5(i)	Position vectors of $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are: $\begin{pmatrix} 0 \\ 7 \\ 11 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}, \begin{pmatrix} 0 \\ 14 \\ 6 \end{pmatrix}$.
(ii)	$\overrightarrow{OQ} = \begin{pmatrix} a/2 \\ 7 \\ 11 \end{pmatrix}$ $\overrightarrow{BQ} = \begin{pmatrix} a/2 \\ 7 \\ 11 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} a/2 \\ 7 \\ 5 \end{pmatrix}$ Normal vector for plane OQS is $\overrightarrow{OQ} \times \overrightarrow{OS} = \begin{pmatrix} a/2 \\ 7 \\ 11 \end{pmatrix} \times \begin{pmatrix} a \\ 14 \\ 0 \end{pmatrix} = \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix}$ Hence length of projection of $\overrightarrow{BQ}$ onto plane OQS is $ \overrightarrow{BQ} \times \frac{\begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix}}{\sqrt{(-154)^2 + (11a)^2}} = \sqrt{90}$ Simplifying, we have $\frac{\sqrt{(55a^2)^2 + 770^2 + (\frac{11}{2}a^2 + 1078)^2}}{\sqrt{154^2 + (11a)^2}} = \sqrt{90}$ $\frac{121}{4}a^4 + 3993a^2 - 379456 = 0$ Using GC, we obtain $a = 14i, -14i, 8$ or -8. Since a is a positive real number, $a = 8$. (Ans) Alternative method : Let F be the foot of perpendicular from B to OQS. Equation of BF is $\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix}$.

$$\overrightarrow{OF} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix}$$

Equation of plane OQS is $r \cdot \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix} = 0$ since plane passes through origin

$$\left[\begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix} \right] \cdot \begin{pmatrix} -154 \\ 11a \\ 0 \end{pmatrix} = 0$$

$$(154)^2 \lambda + 121\lambda a^2 = 0$$

$$\lambda = 0$$

$$\overrightarrow{OF} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} \text{ so } F = B$$

$$\text{Length of projection} = |\overrightarrow{BF}| = |\overrightarrow{BQ}| = \left| \begin{pmatrix} \frac{a}{2} \\ 7 \\ 5 \end{pmatrix} \right| = \sqrt{90}$$

$$a = 8$$

(iii)

$$\overrightarrow{AB} = \begin{pmatrix} 0 \\ -7 \\ -5 \end{pmatrix}, \overrightarrow{AC} = \begin{pmatrix} 0 \\ 7 \\ -5 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 70 \\ 0 \\ 0 \end{pmatrix}$$

Hence a normal vector to plane ABC is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

Since plane DEF is parallel to ABC, the two planes share the same normal vector.

Vector equation of plane **DEF** is

$$r \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$r \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 8. (\text{Ans})$$

6	Advantage Any of the following or reasonable answer: 1. The numbers are truly random and free from bias. 2. Analysis of data is relatively easy. Disadvantage Any of the following or reasonable answer: 1. The sample chosen is not a good representation of the population. 2. As the population is large, it is difficult and time consuming to identify every member of the population. 3. Sometimes, due to certain reasons, it is not possible to get access to some members who have been chosen for the sample. A stratified sample may be a better representation of the Singaporean population which consists of different age groups, ethnic groups, income groups and gender. Possible strata : different age groups or ethnic groups or income groups.
7(a)	(i) Case 1: 2 red + 2 green, no of ways = ${}^5C_2 \times {}^8C_2 = 280$ Case 2: 1 red + 3 green, no of ways = ${}^5C_1 \times {}^8C_3 = 280$ Case 3: 4 green, no of ways = ${}^8C_4 = 70$ Total no of ways = $280 + 280 + 70 = 630$ (ii) No of ways to select at least 1 of each colour = $n(\text{any 4 of 13 balls}) - n(4 \text{ of 5 red}) - n(4 \text{ of 8 green})$ = ${}^{13}C_4 - {}^5C_4 - {}^8C_4$ = 640
7(b)	Case 1: blue trousers, i.e. any 2 of 5 skirts: no of ways = ${}^5P_2 = 20$ Case 2: green trousers, i.e. no green skirt: no of ways = ${}^4P_2 = 12$ Case 3: yellow trousers, i.e. no yellow skirt: no of ways = ${}^4P_2 = 12$ Total no of ways = $20 + 12 + 12 = 44$
8	(i) P (it will take exactly n throws of the biased die to obtain a '6') = P (the first $(n - 1)$ throws are not '6' and the nth throw is a '6') = $(1 - p)^{n-1} p$ (ii) P (it takes exactly 3 throws to obtain a '6') = $(1 - p)^2 p = \frac{9}{64}$ $p^3 - 2p^2 - p - \frac{9}{64} = 0$ $p = 0.25, 0.4243, 1.326$ (from GC-Polynomial solver) $p = 0.4243 \approx 0.424$ since $\frac{1}{4} < p < 1$

	$P(\text{obtain a '5'}) = \frac{1 - 0.4243}{5} = 0.11514$ $P(\text{They obtained the same number} \mid \text{they obtained a number larger than 4}) = \frac{P(\text{they obtained the same number and each obtained a number greater than 4})}{P(\text{they each obtained a number greater than 4})}$ $\frac{P(\text{each obtained 5 or each obtained 6})}{P(\text{each obtained 5 or 6})} = \frac{P(5, 5) + P(6, 6)}{[P(5 \text{ or } 6)]^2}$ $= \frac{(0.11514)^2 + (0.4243)^2}{(0.11514 + 0.4243)^2}$ $= 0.664 \text{ (to 3 sf)}$ Alternative solution : $= \frac{P(\text{each obtained 5 or each obtained 6})}{P(\text{each obtained a number greater than 4})} = \frac{P(5, 5) + P(6, 6)}{P(5, 5) + P(6, 6) + 2P(6, 5)}$ $= \frac{(0.11514)^2 + (0.4243)^2}{(0.11514)^2 + (0.4243)^2 + 2(0.11514)(0.4243)}$
9	(i) $Y = aX + b \sim N(30a + b, 16a^2)$ $P(Y < 4) = P(Y > 16) = 0.06681$ By symmetry (since the two end tailed areas are equal), $30a + b = \frac{4 + 16}{2} = 10 \text{ --- (1)}$ $\frac{4 - 30a - b}{4a} = -1.5000 \Rightarrow 24a + b - 4 = 0 \dots (2)$ From (1), $b = 10 - 30a$ sub. into (2) $24a + 10 - 30a - 4 = 0$ $a = 1$ and $b = -20$ (Shown) Alternative solution : $P\left(Z < \frac{4 - 30a - b}{4a}\right) = 0.06681$ $\frac{4 - 30a - b}{4a} = -1.5000 \Rightarrow 24a + b - 4 = 0 \dots (1)$ $P(Y > 16) = 0.06681 \Rightarrow P(Y < 16) = 0.93319$ $P\left(Z < \frac{16 - 30a - b}{4a}\right) = 0.93319$ $\frac{16 - 30a - b}{4a} = 1.5000 \Rightarrow 36a + b - 16 = 0 \dots (2)$ (ii) $Y = X - 20 \sim N(10, 16) \Rightarrow \bar{Y} \sim N(10, \frac{16}{10})$ $X \sim N(30, 16) \Rightarrow \bar{X} \sim N(30, \frac{16}{10})$ Therefore, $\bar{X} - \bar{Y} \sim N(20, \frac{16}{5})$ P (the difference between $\bar{X}$ and $\bar{Y}$ is less than 15)

	$= P(\bar{X} - \bar{Y} < 15) = P(-15 < \bar{X} - \bar{Y} < 15) = 0.00259$
10(i)	$\sum x = 634.2 \quad \sum x^2 = 5653.5 \quad n = 75$ $\bar{x} = \frac{634.2}{75} \quad s^2 = \frac{1}{74} \left[5653.5 - \frac{634.2^2}{75} \right]$ $= 8.456 \quad = 3.928$ $\approx 8.46 \quad \approx 3.93$
(ii)	$H_0 : \mu = 8$ $H_1 : \mu > 8$ Level of significance, $\alpha = 0.05$ Based on H_0, since $n \geq 50$, by Central Limit Theorem, $\bar{X} \sim N\left(8, \frac{3.928}{75}\right) \text{ approximately}$ By Z-test, p-value = $P(\bar{X} > 8.456)$ = 0.02315 (to 4 sf) Since the p-value = 0.02315 < $\alpha = 0.05$, we reject H_0. There is sufficient evidence at 5% significance level to conclude that the company's claim of 8 hours is an under-estimation of the mean.
(iii)	$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$ Based on H_0, since $n = 75 \geq 50$, by Central Limit Theorem, $\bar{X} \sim N\left(\mu_0, \frac{3.928}{75}\right) \text{ approximately}$ If H_0 is not rejected, p-value is larger than 0.05. $P(\bar{X} > 8.456) > 0.05$ $P\left(Z > \frac{8.456 - \mu_0}{\sqrt{0.05237}}\right) > 0.05$ $P\left(Z < \frac{8.456 - \mu_0}{\sqrt{0.05237}}\right) < 0.95$ $\frac{8.456 - \mu_0}{\sqrt{0.05237}} < 1.6449$ $\mu_0 > 8.08$

11	(i) Product moment correlation coefficient $r = 0.979$. There exists a strong positive linear correlation between x and y. (ii) Equation of least square regression line is $y = 18.5 + 0.564x$. (iii) Given that $y = 30$, $x = 20.4$ from the equation. She left at 7am. This estimate is reliable since $r \approx 1$ so we can use the equation of y on x to estimate x, giving y and $y = 30$ is within the given data range. (2 reasons) (iv) $z = \text{time available} - \text{time taken}$ $= 50 - x - y$ $= (50 - x) - (a + bx)$ $= (50 - x) - (18.5 + 0.564x)$ $= 31.5 - 1.564x$. (v) For $z = 0$, $x = \frac{31.5}{1.564} = 20.14 \approx 20\text{min}$ The latest time when Ms Chan leaves her house is 7 a.m.
12	(i) Let X be the rv “the number of new influenza infections in Fluting in a day” Let Y be the rv “the number of new influenza infections in Healthville in a day” $X \sim \text{Po}(\frac{15}{7})$ and $Y \sim \text{Po}(\frac{12}{7})$ Therefore, $X+Y \sim \text{Po}(\frac{27}{7})$ $P(\text{total number of influenza infections in the two villages is greater than 5 in one day})$ $= P(X + Y > 5) = 1 - P(X + Y \leq 5)$ $= 0.193$ (to 3 sf) (ii) The Additive Property for Poisson variables can be applied if it is assumed that that the number of infections in the two villages is independent of each other. (iii) Let F be the rv “the number of new influenza infections in Fluting in a month” Let H be the rv “the number of new influenza infections in Healthville in a mth.” $F \sim \text{Po}(60)$ and $H \sim \text{Po}(48)$ Since $\lambda = 60 > 5$, $F \sim N(60, 60)$ approximately and $H \sim N(48, 48)$ approximately. Therefore, $H - F \sim N(-12, 108)$ $P(\text{the number of influenza infections in Healthville exceeds the number of infections in Fluting})$ $= P(H > F)$ $= P(H - F > 0) \xrightarrow{cc} P(H - F > 0.5)$ $= 0.115$ (to 3 sf) Let A be the number of patients who die from the disease out of the 40 in hospital.

$A \sim B(40, p)$
 $\text{Var}(A) = 40p(1-p)$. Therefore, $40p(1-p) = 0.784$
 $40p - 40p^2 = 0.784$. From GC, $p = 0.02$ (Shown)

Hence, $A \sim B(n, 0.02)$

It is given that $P(A \geq 1) > 0.5 \Rightarrow P(A = 0) < 0.5$

$$(0.98)^n < 0.5 \Rightarrow n > \frac{\ln 0.5}{\ln 0.98} = 34.3 \text{ (to 3 sf)}$$

Least $n = 35$.

The End